



Ensemble Techniques

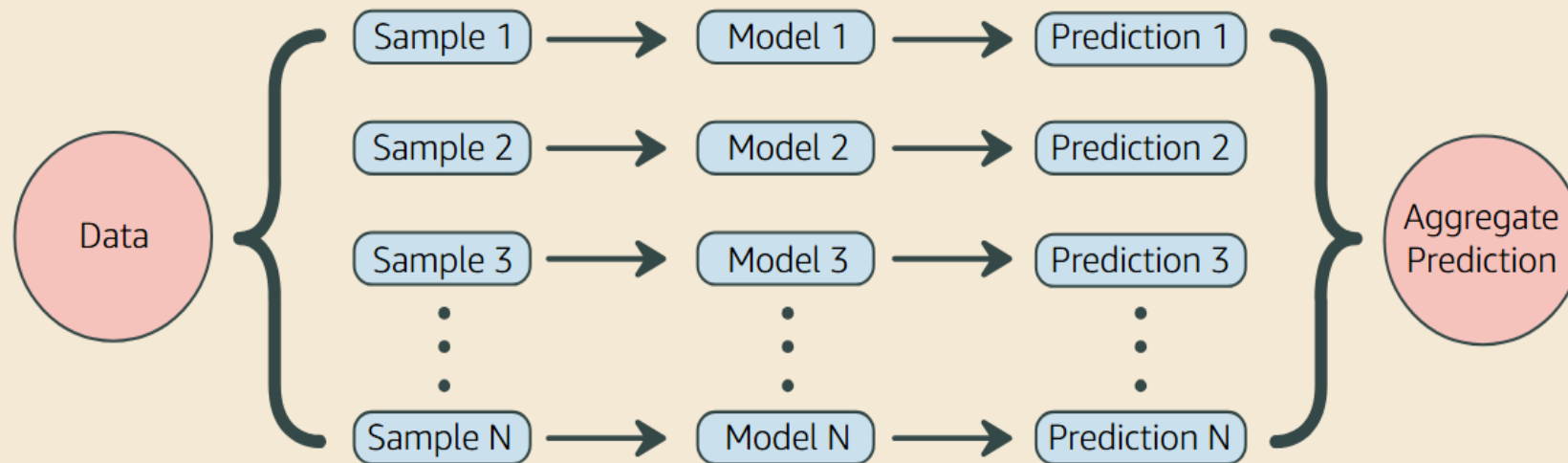
Wisdom of the Crowd

- How many McDonald Store is there in USA?
- How many mid day meals are served in India per day?
- How many total passes attempted in FIFA 2026?



Ensemble Techniques

Ensemble Learning



Bagging and boosting solve different ensemble problems

BAGGING

Build independently

- models train in parallel
- each sees a different sample or feature view
- predictions are averaged or voted
- primary goal: reduce variance

RANDOM FOREST

BOOSTING

Build sequentially

- each model learns after the previous one
- later models focus on residual errors
- contributions are combined, often weighted
- primary goal: reduce bias and improve fit

GRADIENT BOOSTING

Both can improve predictive performance. The right choice depends on data, computation, tuning, and the cost of mistakes.



Random Forest



Random Forest Visualizer

MLU Explainer – Random Forest



Gradient
Descent

UNDERSTANDING GRADIENT DESCENT IN MACHINE LEARNING



Gradient Descent- Case Study

